Ethan Davis

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Research Interests

probabilistic machine learning, uncertainty quantification, Bayesian inference, generative modeling, posterior sampling, Bayesian hypothesis testing, transfer and continual learning, causal inference, spatial and temporal modeling.

Education

M.S., Computer Science 2024–2026
University of Washington — Seattle, WA

B.S., Computer Science 2020–2022
Oregon State University — Corvallis, OR

B.S., Mathematics 2011–2015
University of Portland — Portland, OR

Research Experience

M.S. Thesis 2024–2026
SPANER Lab (Smart Platform for Augmented Neurology Rehabilitation) — University of Washington

- Evaluated Bayesian complete-pooling against frequentist baselines for cross-subject motor-imagery EEG (MI-EEG) classification across twenty datasets and six pairs of classifiers.
- Compared top-performing MI-EEG classifiers, including linear spatial filters, Riemannian geometric filters, and deep neural networks.
- Accumulated effect-size estimates with meta-analysis instruments across overall performance, its calibration and resolution components, discrimination, and sharpness.
- Concluded that complete-pooling effects on calibration and sharpness are statistically but not practically significant, with no detectable difference in performance, resolution, or discrimination.
- Benchmarked with MOABB for standardized preprocessing and model selection across MI-EEG datasets, replacing its built-in statistical evaluation with my own meta-analysis.
- Refined the experiment design after defense, releasing a preprint, software, and raw evaluation outputs, and hypothesizing that partial-pooling would generate larger effects.

Graduate Research 2024–2026
Independent Research — University of Washington

- Studied varying scopes of the machine learning systems stack, several projects beginning as course projects that I extended into independent work.
- Implemented eigendecomposition in C for eigenface image compression with convergence proofs, fusing chained multiplications into single-pass loops to avoid catastrophic cancellation.
- Benchmarked generalized matrix multiplication (GEMM) in NVIDIA CUDA, comparing CPU against GPU and third-party against hand-rolled implementations across matrix sizes.
- Built ant colony optimization for the traveling salesman problem with Hadoop MapReduce on AWS EC2 clusters in Java, generalized as the exploration-exploitation-evaluation (EEE) framework.
- Designed distributed triangle counting with Apache Spark on AWS EC2 clusters in Java for graph analytics and community detection.
- Researched, designed, and built software architectures for end-to-end ML lifecycles with continuous training for ads recommendation systems in social networks.

Publications & Preprints

[1] Ethan Davis. *Bayesian Complete-Pooling in Cross-Subject Classification for Motor Imagery Electroencephalogram*. 2026. arXiv: 2607.22980 [cs.LG]. Under review, Journal of Neural Engineering (submitted July 2026).

[2] Ethan Davis. "Benefits of Being Bayesian: Motor Imagery Electroencephalogram Classification". Master's thesis. University of Washington, 2026. Publication No. 32735698, ProQuest Dissertations & Theses Global.

[3] Ethan Davis. *Exploration-Exploitation-Evaluation (EEE): A Framework for Metaheuristic Algorithms in Combinatorial Optimization*. 2025. arXiv: 2510.05027 [cs.NE].

[4] Ethan Davis. *High Performance Matrix Multiplication*. 2025. arXiv: 2509.04594 [cs.PF].

[5] Ethan Davis. *Linear Algebra for Image Compression*. Technical report. Zenodo, Mar. 2025. DOI: 10.5281/zenodo.17180358.

Presentations & Posters

[1] Ethan Davis. *Benefits of Being Bayesian: Motor Imagery Electroencephalogram Classification (MS Thesis Defense Slides)*. MS thesis defense, University of Washington. Zenodo, June 2026. DOI: 10.5281/zenodo.21724209.

[2] Ethan Davis. *Bayesian Deep Learning for Electroencephalogram Signal Recognition*. Poster presented at PUMPS+AI ACM, Barcelona, Spain. Zenodo, July 2025. DOI: 10.5281/zenodo.21631478.

Data Sets

[1] Ethan Davis. *Bayesian and Frequentist Motor Imagery EEG Classification: Model Scores, Compute Profiling, and MCMC Convergence Diagnostics*. figshare, 2026. DOI: 10.6084/m9.figshare.32176734.

[2] Ethan Davis. *Raw Evaluation Outputs for Bayesian Complete-Pooling in Cross-Subject Classification for Motor Imagery Electroencephalogram*. Zenodo, 2026. DOI: 10.5281/zenodo.21538705.

Software & Reproducibility

[1] Ethan Davis. *Bayesian and Frequentist Pipeline Benchmarking for Cross-Subject Motor Imagery EEG Classification*. Version v2.0-literature. Zenodo, 2026. DOI: 10.5281/zenodo.21540429.

[2] Ethan Davis. *Benefits of Being Bayesian: Motor Imagery EEG Classification (Thesis Software)*. Version v1.0-thesis. Zenodo, 2026. DOI: 10.5281/zenodo.20415596.

[3] Ethan Davis. *Ant Colony Optimization with Hadoop MapReduce on Distributed AWS EC2 Clusters*. Version v1.0.2. Zenodo, Oct. 2025. DOI: 10.5281/zenodo.17274214.

[4] Ethan Davis. *Data Structures and Algorithms: Worked Solutions and Complexity Analyses*. Version v1.0.2. Zenodo, Oct. 2025. DOI: 10.5281/zenodo.17289626.

[5] Ethan Davis. *Eigenfaces for Image Compression: Eigendecomposition in C*. Version v1.0.1. Zenodo, Oct. 2025. DOI: 10.5281/zenodo.17299529.

[6] Ethan Davis. *High Performance Matrix Multiplication: GEMM Benchmarks Across CPU and GPU*. Version v1.0.1. Zenodo, Oct. 2025. DOI: 10.5281/zenodo.17299758.

[7] Ethan Davis. *Triangle Counting with Apache Spark on Distributed AWS EC2 Clusters*. Version v1.0.1. Zenodo, Oct. 2025. DOI: 10.5281/zenodo.17299086.

Teaching Experience

Guest Lecturer February 2026

Data Mining for Machine Learning, graduate course — University of Washington

- Designed and delivered a guest lecture on feature engineering and selection at the invitation of Dr. Erika Parsons.
- Built a runnable Jupyter Notebook on MI-EEG data, letting students execute a full classification pipeline themselves.
- Wrote supplementary proofs of spatial filtering methods, including the common spatial pattern and tangent space mapping, grounded in linear algebra and Riemannian geometric statistics.

Instructional Materials 2025–2026
Self-published — Zenodo

- Released lesson notes on Bayesian ML for MI-EEG spanning eleven sections, from EEG protocols to Gaussian processes and Hamiltonian Monte Carlo, with derivations worked in full.
- Wrote a data structures and algorithms reference spanning two dozen problem categories, pairing worked solutions with complexity analyses.

Lab Documentation 2025–2026
SPANER Lab — University of Washington

- Built the lab's EEG-classification reference from an independent literature review, spanning EEG foundations, machine learning, software, and experiment design.
- Tiered the machine learning material by depth, from introductory texts through applied topics such as physics-informed ML to statistical learning theory and Riemannian geometric statistics.

Teaching Materials

[1] Ethan Davis. *Bayesian Machine Learning for Motor-Imagery EEG: Lesson Notes*. Zenodo, June 2026. DOI: 10.5281/zenodo.21711021.
[2] Ethan Davis. *Data Structures and Algorithms*. Zenodo, 2026. DOI: 10.5281/zenodo.21693224.
[3] Ethan Davis. *Feature Engineering: Lesson Materials*. Zenodo, July 2026. DOI: 10.5281/zenodo.21693577.

Professional Service

Open-Source Contributor 2025–present
MOABB, Mother of all BCI Benchmarks — NeuroTechX

- Documented limitations as GitHub issues and resolved them in pull requests merged upstream, with 12 credited entries shipped in MOABB v1.5.0.
- Contributed fine-grained compute profiling of power, energy, water, and utilization across processors and memory.
- Contributed multi-metric scoring with arbitrary user-defined metrics, and fixes to model pickling and metric reporting.

Package Author 2025–present
tombolo and moabbr — PyPI and Docker Hub

- Released tombolo, a PyPI package exposing an R statistical runtime for hypothesis testing methods most mature in R, containerized as a Docker Hub image for portability.
- Released moabbr, a companion package adapting MOABB to that interface; both packages are pre-1.0 and under active development.

Industry Experience

Software Engineer 2022–2024
SeekOut — Bellevue, WA

- Built ETL pipelines carrying LinkedIn, Google Scholar, and applicant tracking system (ATS) sources into a centralized sink that staged records for search indexing, in C# .NET and Azure.
- Normalized heterogeneous provider schemas into one indexable representation, reconciling the inconsistent fields and duplicate records with which each provider described the same candidate.
- Produced UML diagrams, documentation, and unit tests to enforce maintainable, SOLID software design, supporting scale in both the number of integrations and the throughput of data ingestion.

Software Engineer 2017–2020
Independent Project — Seattle, WA

- Designed and implemented a scalable image-sharing platform using Java Spring micro-services, Docker containers, and Kubernetes orchestration.
- Provisioned AWS infrastructure with Terraform and configuration with Ansible, so the environment could be rebuilt from scratch rather than by hand, eliminating configuration drift.
- Developed a React.js frontend with full CRUD functionality, deployed via Node.js web servers in the Kubernetes cluster, backed by MongoDB and S3 image storage.

Software Engineer 2015–2017
StackBrew — Redmond, WA

- Built a JavaScript AST interpreter displaying variable contents on hover in the browser editor, written in Node.js and rewritten as a C++ addon once interpretation became the bottleneck.
- Built a git-based micro-service enabling version control and automated CI/CD pipeline integration, streamlining development workflows.
- Implemented a collaborative editing micro-service using CRDTs for consistent concurrent edits; platform technologies included Node.js, Go, MongoDB, Docker, Kubernetes, and GCP.

Honors & Awards

Statement of Accomplishment July 2025
PUMPS+AI ACM — Barcelona Supercomputing Center

Dean's List Autumn 2021
College of Engineering — Oregon State University

Updated August 2026